

THEO GUEGAN

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EXPERIENCE

Founding Engineer, Embodied AI

Feb 2026 – Present

Opalin (Claire Delaunay, ex-NVIDIA Robotics)

Bay Area, CA

- Trained visuomotor policies with JEPA architectures: latent-predictive encoders over multi-view video and proprioception, distilled into real-time policies running on physical robots.
- Built variance-targeted data collection and curation, the largest source of model gains: 90 curated demonstrations beat 250 unfiltered ones, folding success 20% to 98%.
- Frozen-DINOv3 phase, progress and contact models pre-annotate every episode at 97% top-1, cutting human labeling 70% and automating curation.
- End-to-end data engine: multi-stream sync, multi-sensor system, 3D dataset viewer (multi-view + URDF) used daily for inspection and QA.
- VLA fine-tuning with flow matching and LoRA/PEFT on multi-GPU; real-time runtime with Pinocchio IK/FK, RNEA, impedance and position control to 0.1 mm over CAN.

Embedded Drone Software Engineer, Intern

Sep 2024 – Feb 2025

Thales Land & Air Systems

Vélizy-Villacoublay, France

- Architected a real-time embedded Lua scripting engine in C++ (TDD), enabling rapid mission prototyping on flight hardware.
- Integrated local LLMs via Rust and Docker under tight compute constraints (85% accuracy); owned the stack from cross-compilation to live demo.

Autonomous Vehicle Control Lead

Feb 2024 – Jun 2025

UTonome — UTAC Challenge

Compiègne, France

- Led sim-to-real deployment of the control stack onto a Renault Zoe (ROS/Python): waypoint control law, adaptive cruise control, obstacle avoidance; 99% safety in simulation.
- Won the 2025 open category and the 2024 best-school award.

RESEARCH

Behavior Cloning of MPC for 3-DOF Robotic Manipulators | arXiv:2606.00383

- Poster at the Workshop on RL in the Era of Imitation Learning, IEEE ICRA 2026.
- Distilled an MPC policy into neural surrogates: 3x lower inference latency at 85% success; static architectures outperform temporal ones.

SKILLS

Learning: self-supervised and latent-predictive (JEPA) objectives, video and multimodal architectures, VLAs, flow matching, imitation learning, LoRA/PEFT, evaluation design

Scale: multi-GPU distributed training, efficient inference, data curation pipelines, dataloaders, PyTorch, JAX, Docker, CI/CD

Planning & control: MPC, model-based planning, Pinocchio IK/FK and RNEA, impedance control, ROS2, dora-rs, RTOS, CAN

Languages: Python, Rust, C++17, C, MATLAB, Bash

Simulation: MuJoCo, Isaac Sim, Genesis, Gazebo

EDUCATION

University of Waterloo

Waterloo, Canada

M.Sc. Computer Engineering, AI/ML

Sep 2025 – Dec 2025

- GPA 4.0/4.0 — Deep Learning (Bryan Tripp), Machine Learning (Terrence C. Stewart), Image Processing.

Université de Technologie de Compiègne

Compiègne, France

M.Sc. Computer Science, Embedded & Autonomous Systems

Sep 2021 – Jun 2026

- GPA 5.0/5.0 — Robotics Control, Embedded Systems, Kinematics.

PROJECTS & AWARDS

Open source: SO-ARM-101 and Reachy Mini manipulation stacks; autonomous drones; peillute, a distributed P2P payment system in Rust; Kaggle Brain-to-Text '25.

Hackathons: 2nd prize, Seed x NVIDIA x Hugging Face Embodied AI (Santa Clara, 2026); 2nd prize, Open World by VLGE (San Francisco, 2026); winner, UTC x ST2I (France); Mistral AI Worldwide (San Francisco, 2026); SWARM Robotics (Canada).